

Quantize PointNet and Deploy it on FPGA Chips for Particle Physics Experiments

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Overview

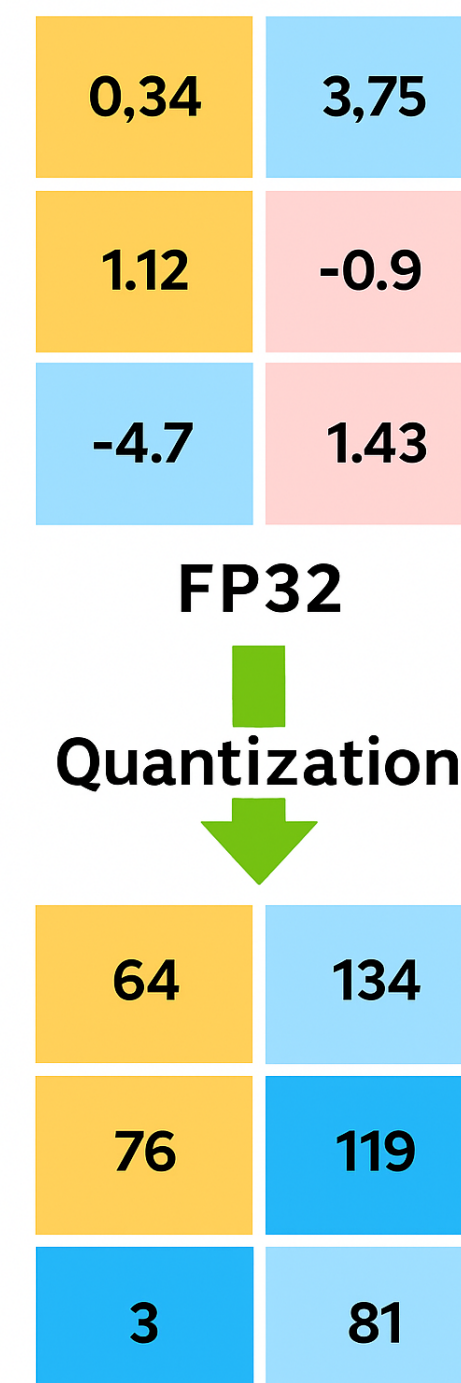
For our project, we aim to quantize a deep learning model to deploy on FPGA chips for particle physics experiments. This endeavor aims to reduce the PointNet model's resource demands, allowing us to deploy it on the Field-Programmable Gate Array chips, which have limited memory resources. Deploying PointNet on FPGA chips will enable us to utilize the model to extract information from the KamLAND-Zen experiment that searches for Neutrinoless Double-Beta Decay and other particle detectors alike.

Key Ideas

Quantization - Quantization reduces the numerical precision of neural networks, enabling PointNet to run efficiently on memory-constrained FPGA hardware without significantly sacrificing accuracy.

PointNet - PointNet is a deep learning model designed for unordered point cloud data. In our project, it learns to predict the 3D location and energy of physics events from detector signals.

FPGA - FPGAs are reprogrammable chips ideal for real-time, low-latency processing. We deploy a quantized version of PointNet on an FPGA to perform fast, on-device event reconstruction for particle physics experiments.



Dataset

We used simulated detector data from the **KamLAND-Zen experiment**, which searches for **neutrinoless double-beta decay**. Each event is captured as a *point cloud* from 2,126 photomultiplier tubes (PMTs) arranged in a spherical detector.

Per-PMT Features:

1. 📍 **x, y, z coordinates** – PMT's location in 3D space
2. ⌚ **Arrival time** – when the photon hit the PMT
3. 💡 **Integrated charge** – light intensity detected
4. 🚩 **Trigger flag** – 1 if light was detected, 0 otherwise

This results in a 2126×6 input matrix for each event.

Events	PMTs per Event	Features per PMT
100,000	2,126	6 (x, y, z, time, charge, trigger)

Table 1. KamLAND-Zen Dataset Statistics

Input Shape	Output Labels
2126×6 (point cloud)	x, y, z position + energy

Table 2. PointNet Input/Output Mapping

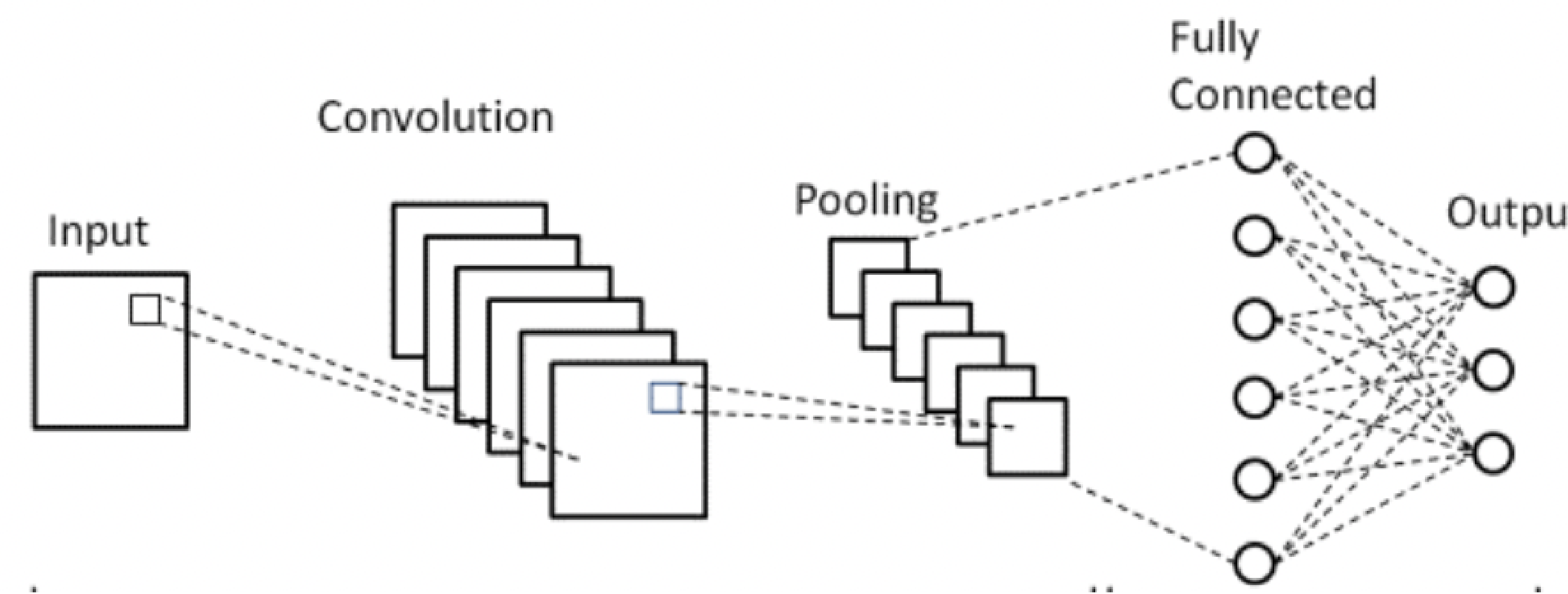
This labeled simulation data is used to train and validate our quantized PointNet model for accurate and efficient reconstruction of physics events on an FPGA.

Methods

We created a general-purpose framework using **Optuna** to optimize the architecture and quantization parameters of any neural network model. Our goal was to quantize PointNet and **minimize an objective function** defined as:

$$\text{Objective} = \text{model parameters} \times \text{loss}$$

Our baseline model was a **6-layer neural network** designed to map detector point cloud data to event position and energy. The network consisted of three convolution layers, followed by a multi-layer perceptron (MLP) with two fully connected layers.



After developing our baseline model, we designed two additional PointNet variants, each with unique layer configurations and optimized hyperparameters. Our goal was to identify the model that best minimized our objective function while remaining lightweight enough for deployment on a resource-constrained FPGA chip.

We quantized the model using **QKeras** by adjusting the number of **system bits (sys_bits)** to reduce memory and power usage. We then used our **custom framework built on top of Optuna** to perform a hyperparameter search. This framework automated the optimization of model architecture and quantization settings for any given model. In our case, it helped guide the search for:

- **Latent units** in the MLP layers
- **Kernel sizes** in the convolutional layers
- **System bit precision** for quantization

Model Variants We evaluated three model configurations:

- **Baseline:** CNN → CNN → (CNN + MaxPool) → MLP → MLP
Parameter Count: 234,183
- **Model 1:** CNN → CNN → (CNN + MaxPool) → MLP
Parameter Count: 238,535
- **Model 2:** (CNN + MaxPool) → MLP
Parameter Count: 8,069

Each model was quantized and tested for its trade-off between inference accuracy and size. Model 2, with the fewest parameters, is the most hardware-efficient; Model 1 and the Baseline are more expressive but more resource-intensive.

Results

We evaluated three models using our quantized neural network framework. While all performed similarly in terms of validation loss, **Model 2** offered the best trade-off between accuracy and parameter count, minimizing the FPGA-suitable objective function:

Model performance across the KamLAND-Zen Simulation Dataset:

Model	Params	Val. MSE	σ_x	σ_y	σ_z	σ_E
Baseline	234,183	913.28	33.70	34.08	36.13	0.06
Model 1	238,535	927.58	33.98	34.60	36.71	0.09
Model 2	8,069	1145.57	36.99	37.20	39.78	0.06

Table 3. Model Comparison: Validation Error and Prediction Spread (σ = standard deviation of prediction error)

Model 2 Final Optimized Hyperparameters (via Optuna):

Hyperparameter	Model 2 Setting
Convolution Filters	128
Kernel Size	3
System Bits (activation)	x=16, k=16, b=16

Table 4. Model 2 Hyperparameters

Model 2 exhibited slightly higher variance across spatial predictions but remained within acceptable thresholds for physics analysis. Its extremely compact architecture (only 8,069 parameters) makes it highly suitable for real-time deployment on resource-constrained FPGA hardware.

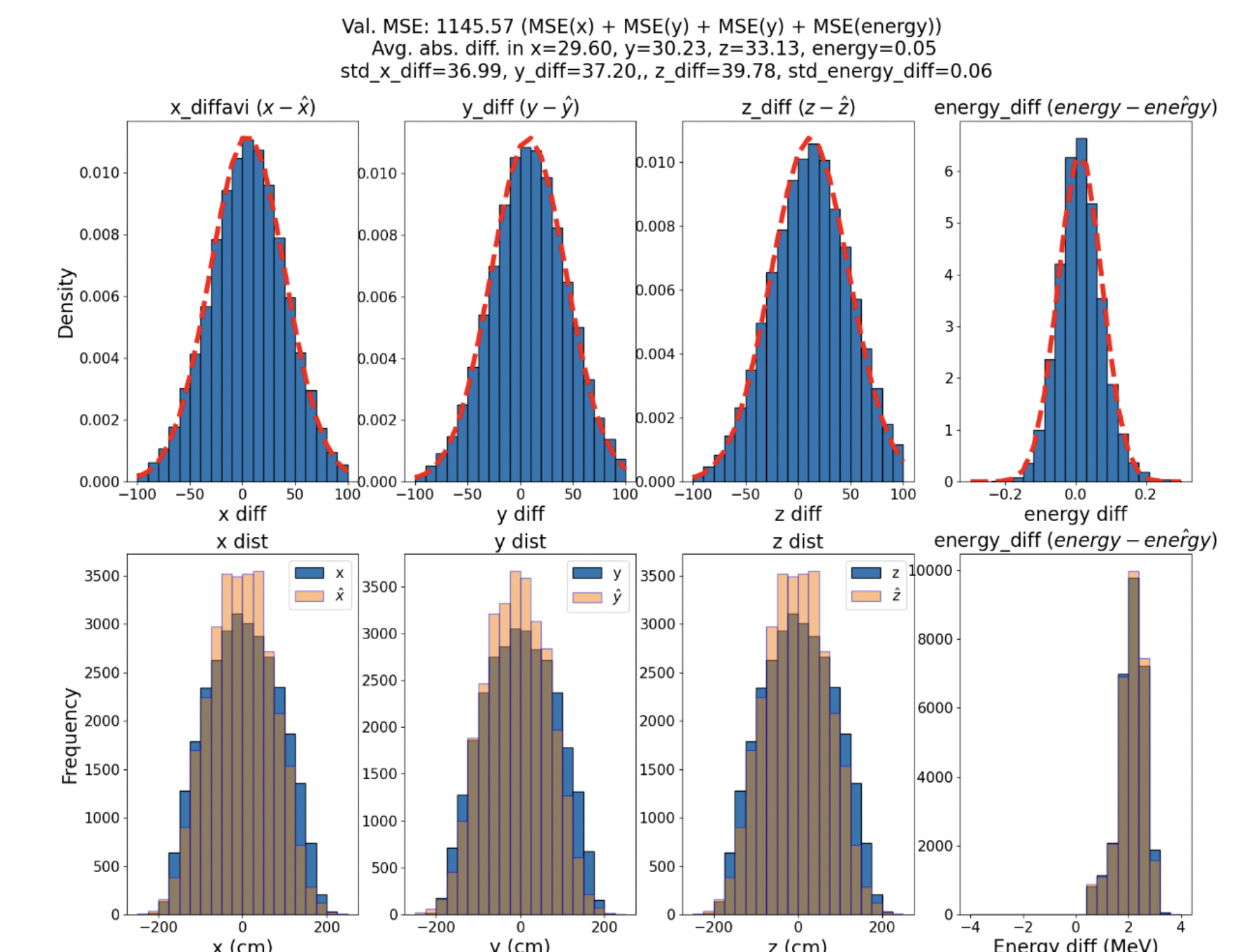


Figure: Model 2 — Lowest objective value and smallest parameter count.

FPGA Deployment

While our goal was to deploy the quantized models directly onto an FPGA, we were unable to complete deployment due to time and hardware limitations. However, based on Model 2's drastically reduced parameter count and lightweight architecture, we anticipate that its inference time on FPGA hardware would decrease significantly—potentially to under **5 milliseconds per event**. This would represent a major step toward real-time event reconstruction in resource-constrained chips!.

Acknowledgments

We would like to extend our sincere thanks to our PhD mentor, **Alex Migala**, and our principal investigator, **Professor Aobo Li**, for their continued support and guidance throughout this project. Under their mentorship, we gained valuable experience at the intersection of machine learning and experimental physics. We are incredibly grateful for the opportunity to contribute to this work and hope to further develop our results into a publication.

We learned so much about physics and ML along the way! 🧠🚀